

Feasibility Viability

Expanded feasibility, risk and deployment analysis for Slide 4. The original slide identifies data availability, existing records, no new hardware, ML tooling, operational risks, geological limitations, domain misalignment, data imbalance, rule-based estimates, active learning and pilot rollout. [\[filecite\turn1file0\L30-L32\]](#)

1. Data feasibility

MOSDAC provides an exact, documented rainfall product rather than only a portal homepage. GSMap_ISRO Rain is available from March 2000 onward, at 0.1° spatial and 1-hour temporal resolution, with IMD gauge adjustment. [\[cite\turn0search0\]](#)

GSI's mineral-hunt portal explicitly lists downloadable Geological Data, Geochemical Data, Geophysical Data, Exploration Reports and Data, Remote Sensing Data and Additional Reports. [\[cite\turn0search11\]](#)

Therefore the pilot can begin with existing public/authorized data and does not need to assume a new satellite or sensor deployment.

2. Geological feasibility

Satellite observations cannot replace subsurface sampling. GSI investigation records show the kind of borehole and assay variability that makes geological ground truth essential; one GSI mineral-investigation record, for example, reports manganese ore zones intersected by boreholes and grade variation within the investigated area. [\[cite\turn0search49\]](#)

NAKSHATRA-X should therefore use satellite signals to prioritize, contextualize and monitor rather than claim direct proof of hidden ore.

3. Operational feasibility

The system can start in shadow mode: it generates predictions and recommendations but does not control production. Mine planners compare its output with existing decisions.

After historical and shadow-mode validation, selected low-risk recommendations can enter the normal approval workflow. Safety-critical control remains under established mine control systems.

4. ML risks and controls

Domain misalignment: geological relationships learned in one deposit may not transfer to another. Control with mine-specific calibration and domain adaptation.

Data imbalance: confirmed high-grade mineralization may be much rarer than background samples. Control with appropriate sampling, class weighting and spatial validation.

False positives: require uncertainty thresholds and human geological review.

Data drift: monitor model performance as mining moves through different geological zones or seasons.

5. Infrastructure viability

The original slide proposes standardized APIs, active learning and a single-mine pilot. These are sensible deployment controls because they reduce integration risk and allow the model to learn from expert feedback.

An offline-first or local-cache mode is useful where mine connectivity is intermittent, but its accuracy and fallback behaviour must be tested separately from the cloud system.

6. Pilot rollout

Phase 1: historical data ingestion and data-quality audit.

Phase 2: retrospective reserve/risk/forecast validation.

Phase 3: live shadow mode with no automated actuation.

Phase 4: human-approved operational recommendations.

Phase 5: multi-mine expansion after model-transfer validation.

7. Feasibility decision criteria

Proceed to scale only if the pilot demonstrates measurable improvement in at least one exploration KPI and one operational KPI without unacceptable false alerts or safety issues.

Evidence and exact source paths

ID	Source	Exact resource / URL	Use in NAKSHATRA-X
S1	MOIL 63rd Annual Report 2024-25	https://moil.nic.in/userfiles/Annual_Report_2024_25.pdf	Reserves/resources, exploration drilling, production and strategic target.
S2	MOIL MD&A; / Annual Report copy	https://moil.nic.in/userfiles/file/InvRel/Financials/Annual_Report_2024-25.pdf	FY25 manganese production and import context.
S3	MOSDAC — GSMap ISRO Rain	https://www.mosdac.gov.in/gsmmap-isro-rain	Exact rainfall product; hourly, 0.1° and March 2000 onward.
S4	GSI Reports database	https://gsi.gov.in/gsiReport/	Official geological/exploration report search.
S5	GSI Mineral Hunt data portal	https://hackathon.gsi.gov.in/login	Direct geology, exploration and remote-sensing data routes.
S6	GSI geology data	https://nis.bisag-n.gov.in/download/Geology/	Direct geology data directory.
S7	GSI exploration reports/data	https://nis.bisag-n.gov.in/download/Exploration_data_reports/	Direct exploration data/report directory.
S8	GSI remote sensing ASTER	https://nis.bisag-n.gov.in/download/Remotesensing_ASTER/	Direct remote-sensing directory.
S9	GSI/OGD manganese deposit dataset	https://tn.data.gov.in/catalog/location-manganese-ore-deposits-in-dia-and-its-salient-features	Specific manganese-deposit location/geology dataset.
S10	Indian Minerals Yearbook portal	https://ibm.gov.in/IBMPortal/pages/Indian_Minerals_Yearbook	Official IBM source for manganese statistics.
S11	Indian Minerals Yearbook 2024	https://ibm.gov.in/writereaddata/files/177426215469c1178a48453IMYB_2024_EBookFinal.pdf	Producer/location and manganese industry data.
S12	ESA — Sentinel-1 seeing through clouds	https://www.esa.int/ESA_Multimedia/Videos/2014/03/Sentinel-1_seeing_through_clouds	SAR capability through cloud/rain and day/night.
S13	ESA — Introducing Sentinel-1	https://www.esa.int/Applications/Observing_the_Earth/Copernicus/Sentinel-1/Introducing_Sentinel-1	C-band SAR, all-weather/day-night mission context.